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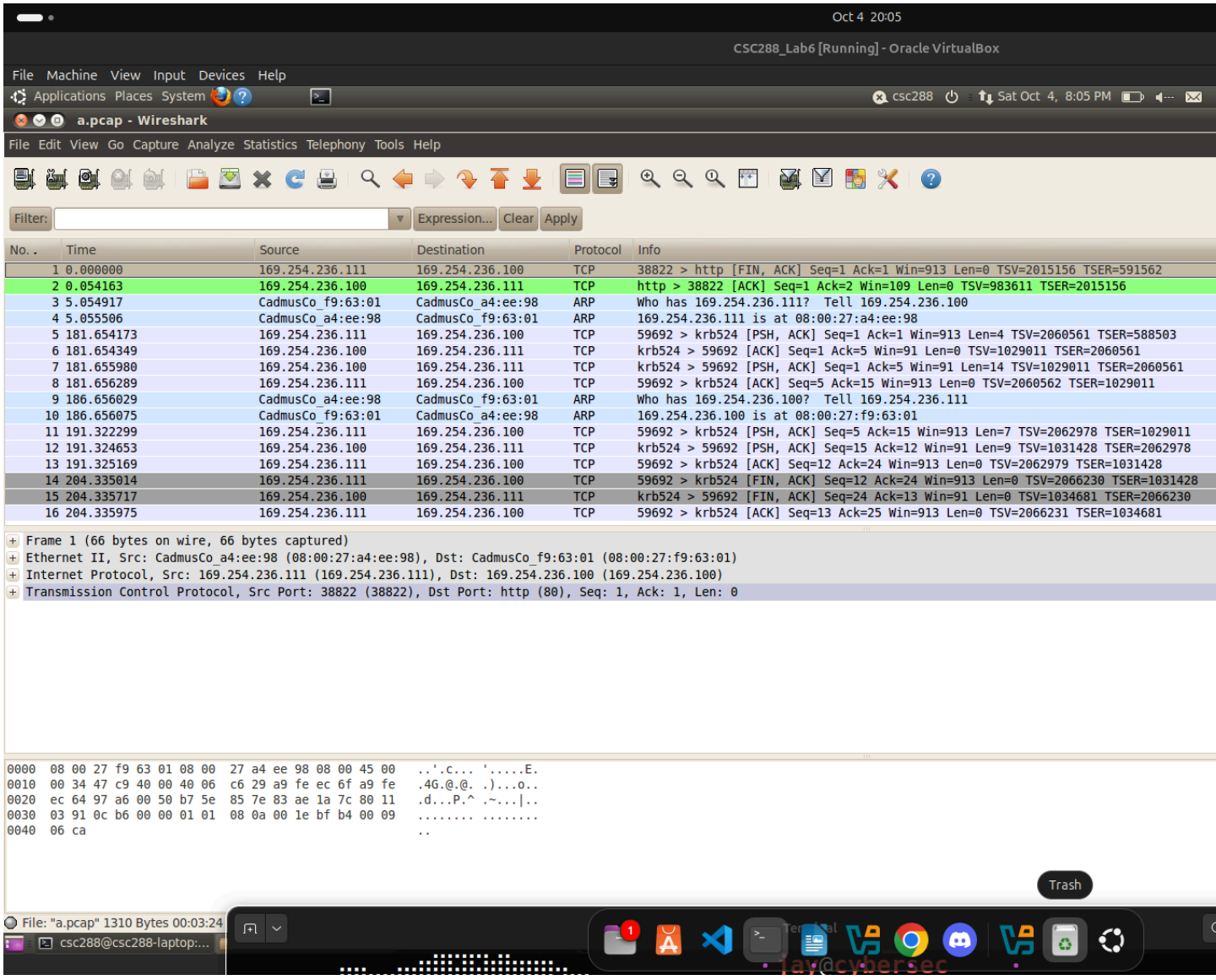
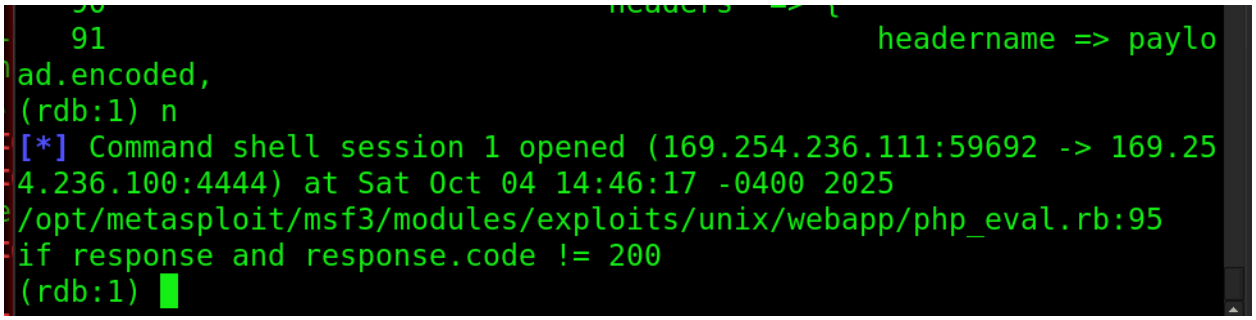
Section 1: Network traffic analysis

Summary: Wireshark captures reveal that 169.254.236.111:59692 initiated a TCP connection to 169.254.236.100:4444, where the target is listening.

Evidence & description:



- **Figure 1:** Wireshark capture (packet list overview).



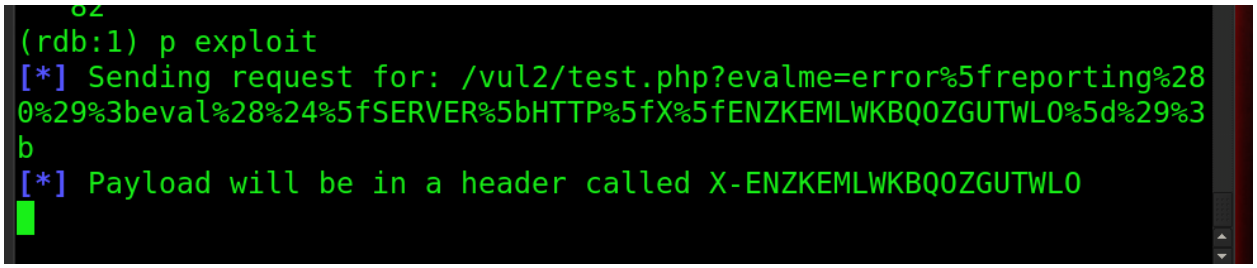
- **Figure 2:** Metasploit console: session opened (169.254.236.111:59692 -> 169.254.236.100:4444).

Analysis: The TCP three-way handshake confirms 169.254.236.111 actively connected to the listening port 4444 on the target system.

Section 2: Exploit execution flow inside Metasploit

Summary: The exploit injects a bind shell payload via a custom HTTP header (X-ENZKEMLWKBQOZGUTWL0). After delivery, exploit_driver.rb calls payload.wait_for_session() at line 232 to wait for the target to establish the bind listener.

Key debugger snapshots & interpretation:



- **Figure 3:** HTTP request showing payload header.

```
[220, 230] in /opt/metasploit/msf3/lib/msf/core/exploit_driver.rb
   220
   221                          # Log a more verbose version
   222                          elog("Exploit exception (#{exploit.
refname}): #{e.class}: #{e}", 'core', LEV_0)
   223                          dlog("Call stack:\n#{$@.join("\n")}
", 'core', LEV_3)
   224                  end
   225
   226                  # Wait the payload to acquire a session if
this isn't a passive-style
   227                  # exploit.
=> 228                  return if not delay
   229
   230                  if (force_wait_for_session == true) or
(rdb:1) p delay
2
(rdb:1)
```

- **Figure 4:** exploit_driver.rb:244 (exploit, payload = ctx).

```
   229
   230                  if (force_wait_for_session == true) (
   231                          (exploit.passive? == false an
oit.handler_enabled?)
=> 232                          begin
   233                                  self.session = payloa
 for session(delay)
```

- **Figure 5:** Delay check (line ~228).

```
   229
   230                  if (force_wait_for_session == true) (
   231                          (exploit.passive? == false an
oit.handler_enabled?)
=> 232                          begin
   233                                  self.session = payloa
 for session(delay)
```

- **Figure 6:** self.session = payload.wait_for_session (line ~232).

Analysis: The driver waits for the payload to bind port 4444, then the attacker connects to it.

Section 3: Tracing into handle_connection()

Summary: Once the connection is established, the framework routes it to a handler. The handler() method retrieves the socket via self.client.conn (line 206-207), which contains RHOST=169.254.236.100 and LPORT=4444. A BindTcpHandlerSession thread spawns and calls handle_connection() to wrap the socket into a session obj

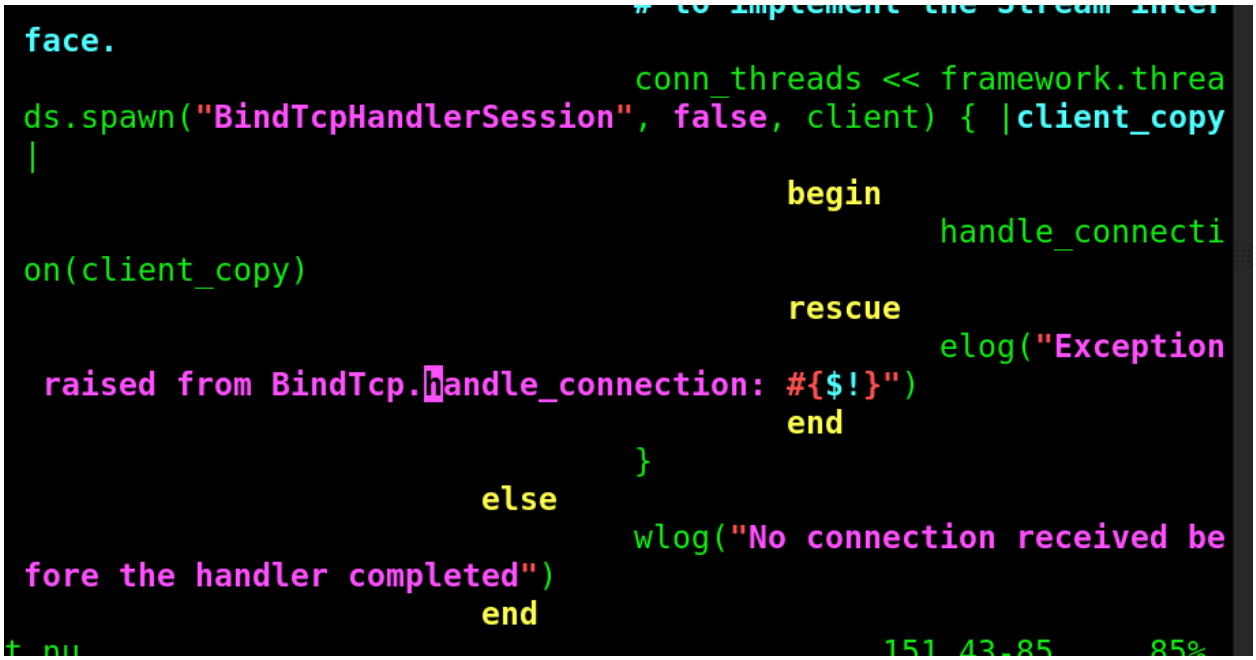
Debugger snapshots & interpretation:

```
   204          def handler(nsock = nil)
   205                  # If no socket was provided, try the global
 one.
=> 206                  if ((!nsock) and (self.client))
   207                          nsock = self.client.conn
   208                  end
   209
   210                  # If the parent claims the socket associate
d with the HTTP client, then
(rdb:1) p handler
nil
(rdb:1) step
/opt/metasploit/msf3/lib/msf/core/exploit/http/client.rb:206
if ((!nsock) and (self.client))
(rdb:1) p nsock
nil
```

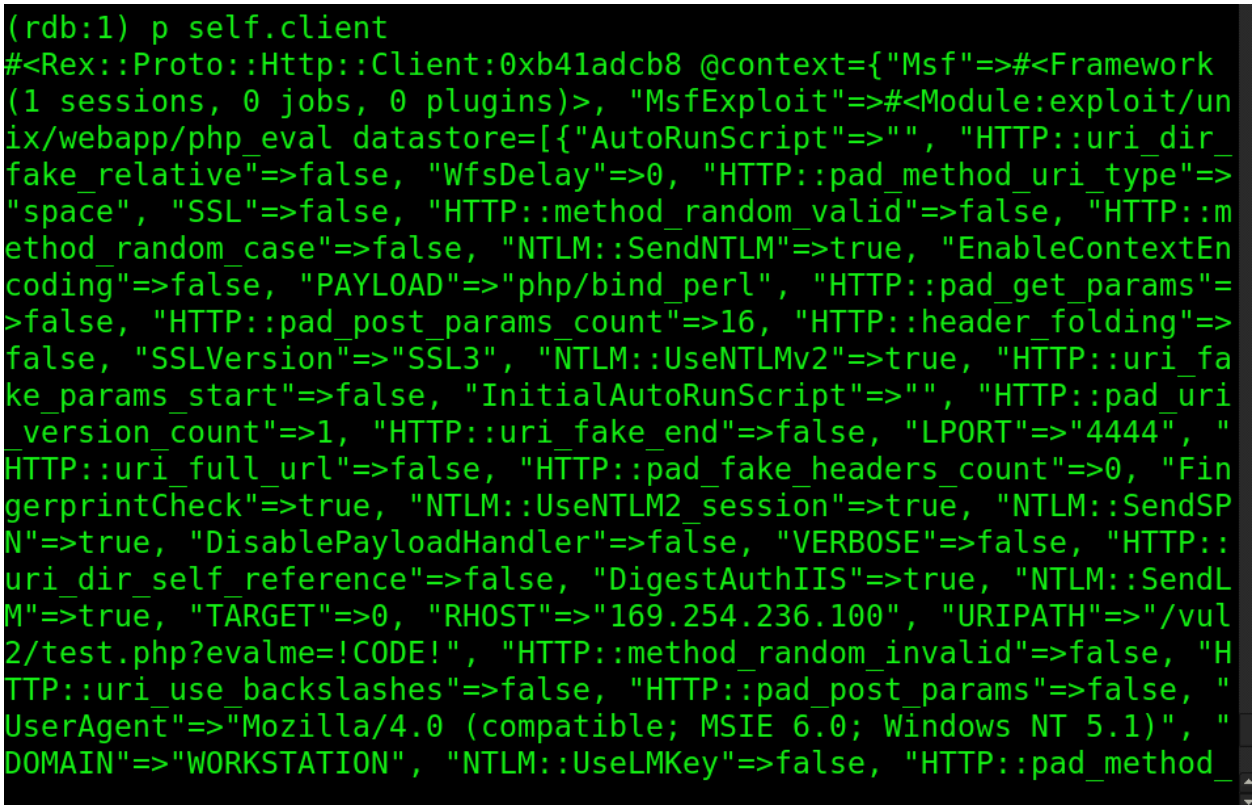
- **Figure 7:** handler method definition (line ~204).

```
   201          # Passes the client connection down to the handler
to see if it's of any
   202          # use.
   203          #
   204          def handler(nsock = nil)
   205                  # If no socket was provided, try the global
 one.
=> 206                  if ((!nsock) and (self.client))
   207                          nsock = self.client.conn
   208                  end
   209
   210                  # If the parent claims the socket associate
d with the HTTP client, then
(rdb:1) n
/opt/metasploit/msf3/lib/msf/core/exploit/http/client.rb:207
nsock = self.client.conn
```

- **Figure 8:** nsock = self.client.conn (line ~206).



- **Figure 9:** BindTcpHandlerSession spawn & handle_connection() invocation.



- **Figure 10:** p self.client showing RHOST and LPORT values.

Flow summary: The handle_connection() method manages I/O between the console and remote shell, completing session creation.

Section 4: Conclusion & who starts the connection to port 4444

Who initiated the connection: Based on the combined evidence (Wireshark packets + Metasploit console message + client metadata), the initiating side is **169.254.236.111:59692** connecting to **169.254.236.100:4444**. The target (169.254.236.100) is the side listening on port 4444 (bind), and the attacker/ephemeral-host (169.254.236.111) actively created the TCP connection to it. In short: **the attacker machine initiated the TCP connection from ephemeral port 59692 to the target's port 4444**.

How the session is created (one-line summary): the exploit injects a payload (HTTP header), the payload triggers a bind/listen on port 4444 on the target, the attacker opens a TCP connection to that port, the framework's handler picks up the socket via self.client.conn and handle_connection() then wraps that socket in a session object (self.session = payload.wait_for_session), and Metasploit reports **"Command shell session opened"**.